Standard Flips of Discrepancy One: Extremal J-Normalization and the Meijer Aperture at $\nu = 1$

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Abstract

Shen and Shoemaker compute the extremal quantum spectrum of a standard flip $X \dashrightarrow X'$ with exceptional locus $\mathbf{P}(V) \subset X$, $\operatorname{rank} V = r$, $\operatorname{rank} V' = s$, and show that the Gamma-class decomposition of $H^*(X)$ attached to the Belmans–Fu–Raedschelders semiorthogonal decomposition is a decomposition into asymptotic classes. Their Theorem 4.4, which identifies an explicit hypergeometric series with the extremal J-function of the local model, assumes $r - s > 1$; their Remark 4.5(3) asserts that for $r - s \leq 1$ the series is not J-normalized; and the Barnes asymptotic expansion of their Section 7 is applied under the same inequality. The printed dependency chain of their Theorems 1.2, 1.4 and 9.14 therefore does not reach the discrepancy-one range $r = s + 1$, $s \geq 1$, which contains every codimension-two blow-up.

We supply the two missing steps. First, the degree-d summand of their formula (35) has z-order at most $1 - s - (r - s)d$; for $r = s + 1$ and $s \geq 1$ this is at most -1 for every $d \geq 1$, so the series is $z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$, no mirror-map correction arises, and the cone membership recorded in Remark 4.5(3) identifies the series with the extremal J-function. The only remaining formal failure of that normalization is the degenerate endpoint $s = 0$, whose point fibres contain no extremal line. Second, at $\nu := r - s = 1$ the sector printed after their Lemma 7.4 is unavailable, being the $\epsilon = 1$ case of their own Theorem A.1; the correct $\epsilon = 1/2$ sector still meets the sector of their Proposition 8.2 in an open sector of opening 2π that contains both the nonzero-eigenvalue ray and the tame ray. With the reductions of their Sections 9.1–9.4 unchanged, this extends Theorems 1.2, 1.4, 9.14 and Corollary 1.5 to every standard flip with $r = s + 1$ and $s \geq 1$.

Keywords. standard flip; quantum spectrum; Gamma class; asymptotic class; Meijer G-function; semiorthogonal decomposition.

1 Introduction

A *standard flip* $X \dashrightarrow X'$, in the setting of Bondal–Orlov and of Belmans, Fu and Raedschelders [3], is built from the following data. There is a birational morphism $p\colon X \to \bar{X}$ of smooth projective varieties, an isomorphism away from $F = p^{-1}(Z)$ for a smooth $Z \subset \bar{X}$, and vector bundles $V, V' \to Z$ of ranks r and s with

$$F \cong \mathbf{P}(V), \qquad N_{F|X} \cong \pi^*(V') \otimes \mathcal{O}_{\mathbf{P}(V)}(-1),$$

where $\pi\colon F \to Z$. Blowing up F and contracting the exceptional divisor $E \cong \mathbf{P}(V) \times_Z \mathbf{P}(V')$ in the other direction produces X', which contains $F' \cong \mathbf{P}(V')$ of codimension r.

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Throughout, $r > s$. Since $F \subset X$ has codimension s and $F' \subset X'$ has codimension r, the two blow-up formulas for the canonical class on $\widehat{X} = \mathrm{Bl}_F X$ give $\tau^* K_X - \tau'^* K_{X'} = (r-s)E$. We call

$$\nu := r - s$$

the *discrepancy* of the flip; $s = 1$ is the blow-up of $\bar{X}$ along Z, and $s = 0$ is the degenerate case $X = F = \mathbf{P}(V)$, $X' = \varnothing$, of a projective bundle.

Shen and Shoemaker [13] compute the spectrum of quantum multiplication by $c_1(X)$ after restricting the Novikov parameter to the extremal ray contracted by p, and prove that the Gamma-class decomposition of $H^*(X)$ induced by the semiorthogonal decomposition of $D^b(X)$ of [3] is a decomposition into strong asymptotic classes in the sense of Galkin–Golyshev–Iritani [7]. This is [13, Theorems 1.2, 1.4 and 9.14, Corollary 1.5], and it is stated for standard flips with the explicit remark that it covers the blow-up case $s = 1$ and the projective-bundle case $s = 0$.

The proof passes through an explicit hypergeometric series. In the local model $T = \mathrm{tot}(\pi^* V' \otimes \mathcal{O}_{\mathbf{P}(V)}(-1))$, after the extremal specialization [13, (33)] of Novikov and parameter variables, [13, Theorem 4.4] states that the extremal J-function of T is

$$J^T(q,t,z) = z\, e^{t/z} q^{c_1(T)/z} \sum_{d \geq 0} q^{(r-s)d} \frac{\prod_{j=1}^{s} \prod_{m=0}^{d-1} (\sigma_j - H - mz)}{\prod_{i=1}^{r} \prod_{m=1}^{d} (\rho_i + H + mz)}, \tag{1}$$

which is [13, (35)]. Everything after Section 4 of [13] runs on this formula: the ring presentation [13, (37)] is deduced from it, the extremal spectrum is read off from that presentation, and the modified J-function of [13, (39)–(40)] used in the asymptotic analysis of Sections 7–9 is defined from it.

Three hypotheses in the printed argument constrain ν. [13, Theorem 4.4] is stated under $r - s > 1$. [13, Remark 4.5(3)] asserts that for $r - s \leq 1$ the right-hand side of (1) is not of the form $z\mathbf{1} +$ (lower order in z) and therefore cannot equal the J-function of T, although it does lie on Givental's Lagrangian cone of T and so is an I-function. Finally, the Barnes asymptotic expansion of the Meijer G-function of [13, Lemma 7.4] is introduced by the sentence "For $r - s > 1$", and the sector [13, (64)] printed with it is derived under that inequality. Under the standing assumption $r > s$, the range excluded by these three hypotheses is exactly $\nu = 1$, that is

$$r = s + 1.$$

It contains every codimension-two blow-up, the case $(r,s) = (2,1)$. So the dependency chain printed for [13, Theorems 1.2, 1.4 and 9.14] does not literally reach the discrepancy-one range, even though the statements are made for standard flips and blow-ups without a restriction on ν.

This note supplies the two missing steps. Both are internal to [13]: the first uses only (1) and the cone membership already recorded in [13, Remark 4.5(3)], and the second uses only [13, Theorem A.1], in the form the authors state it.

Theorem 1.1. *Let $r = s + 1$ with $s \geq 1$. Then, in the extremal specialization [13, (33)] and with the same formal variables and cohomological completion as in [13, Theorem 4.4], the degree-d summand of* (1) *has z-order at most -1 for every $d \geq 1$. Consequently* (1) *equals $z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$ with $\bar{\mathbf{t}} = t + \ln(q) c_1(T)$, and equals the extremal J-function of T. In particular the presentation [13, (37)] of [13, Theorem 4.6] holds for $r = s + 1$, $s \geq 1$, as does [13, Theorem 1.2], with the single nonzero eigenvalue $\lambda_0(q) = (-1)^s q$ of multiplicity* $\dim H^*(Z)$.

The negative assertion of [13, Remark 4.5(3)] is therefore accurate, in the range $r > s$ it addresses, only for $s = 0$, where $(r, s) = (1, 0)$ and the degree-one summand of (1) does contribute in z-order zero.

Theorem 1.2. *Let $r = s+1$ with $s \geq 1$, and let $0 \leq k \leq 1$. The asymptotic expansion of [13, Proposition 7.5] for the component indexed by $m = -k$ is valid on $\left|\arg(z/q) + (2m+s)\pi\right| < \frac{3\pi}{2}$, and not on the wider sector obtained by setting $r - s = 1$ in [13, (64)]. The expansion of [13, Proposition 8.2] is valid on $\left|\arg(z/q) - (1-s)\pi\right| < \frac{3\pi}{2}$. These two sectors meet in an open sector of opening 2π containing both the eigenvalue ray $\arg(z/q) = -(2m+s)\pi$ and the tame ray $\arg(z/q) = (1-s)\pi$.*

For a codimension-two blow-up, $(r, s) = (2, 1)$ and $k = m = 0$, the two sectors are $-\frac{5\pi}{2} < \arg(z/q) < \frac{\pi}{2}$ and $|\arg(z/q)| < \frac{3\pi}{2}$, and they meet in $-\frac{3\pi}{2} < \arg(z/q) < \frac{\pi}{2}$, which contains the tame ray 0 and the eigenvalue ray $-\pi$.

Corollary 1.3. *[13, Theorems 1.2, 1.4, 9.9 and 9.14] and [13, Corollary 1.5] hold for every standard flip with $r = s + 1$ and $s \geq 1$, in particular for every codimension-two blow-up.*

The mechanism behind Theorem 1.1 is a single count, and it is worth saying where the natural first count goes wrong. The numerator of the degree-d summand of (1) has sd factors and the denominator has rd factors. Treating all of them as linear in z gives $1 + sd - rd = 1 - \nu d$ for the maximal z-order, which is 0 at $d = 1$ when $\nu = 1$: a curve-induced contribution in z-order zero, hence a nontrivial mirror map, hence exactly the failure asserted in [13, Remark 4.5(3)]. But the inner index in the numerator runs over $0 \leq m \leq d - 1$, so s of its factors are $\sigma_j - H$, free of z. The correct maximal order is $1 + s(d-1) - rd = 1 - s - \nu d$, and at $\nu = 1$ the s missing powers of z are precisely what pushes every $d \geq 1$ summand below z^0 as soon as $s \geq 1$. Once no z-order-zero contribution survives, the z^0 coefficient of (1) is the parameter $\bar{\mathbf{t}}$ itself, the mirror map is the identity, and uniqueness of the J-slice of the cone converts the cone membership of [13, Remark 4.5(3)] into an identification.

Theorem 1.2 is not a strengthening but a correction of an aperture. The Meijer G-function of [13, Lemma 7.4] has $\nu = r - s$, and [13, Theorem A.1] — Barnes' expansion in the form of Meijer [10] and Luke [9] — is valid on $|\arg t| < (\nu + \epsilon)\pi$ with $\epsilon = 1$ for $\nu > 1$ but only $\epsilon = \frac{1}{2}$ for $\nu = 1$. The sector [13, (64)] is the $\epsilon = 1$ sector; substituting $r - s = 1$ into it claims an aperture $\pi/2$ wider on each side than Barnes' expansion supports. The point of Theorem 1.2 is that the smaller, correct sector is still large enough: the common sector with [13, Proposition 8.2] retains opening 2π, so the oriented comparison of the nonzero-eigenvalue and tame expansions that the argument of [13, Section 9] performs is available at $\nu = 1$ as it is for $\nu > 1$.

What is not covered is the formal endpoint $(r, s) = (1, 0)$, the rank-one projective bundle, where $\mathbf{P}(V) = Z$, $X' = \varnothing$, and the point fibres contain no line class $[L]$. Its displayed hypergeometric expression also fails the required order count. Nothing here bears on $\nu > 1$, where the argument of [13] already applies, and nothing here changes any statement of [13] outside the two hypotheses named above.

Section 2 fixes notation and records the three places where ν is constrained in [13], together with what depends on each. Section 3 proves Theorem 1.1, Section 4 proves Theorem 1.2, and Section 5 assembles Corollary 1.3 and states the boundary of the repair.

Convention. Numbered items of [13] are always cited explicitly, as in [13, Theorem 4.4] or [13, (35)], and refer to the version arXiv:2502.08762v2. Displays numbered here, such as (1), are our own.

2 The extremal setting and the three constraints on ν

We keep the notation of [13]. Let $H = c_1(\mathcal{O}_{\mathbf{P}(V)}(1))$, let $\rho_1, \ldots, \rho_r$ and $\sigma_1, \ldots, \sigma_s$ be the Chern roots of V and V' pulled back to $\mathbf{P}(V)$, and let $T = \mathrm{tot}(\pi^* V' \otimes \mathcal{O}_{\mathbf{P}(V)}(-1))$ be the local model, so that $c_1(T) = (r-s)H + c_1(Z) + c_1(V) + c_1(V')$ by [13, (32)].

The J-function of a smooth projective Y is $J^Y(Q,t,z) = zS^Y(Q,t,z)^{-1}\mathbf{1}$ as in [13, (20)]; it satisfies

$$J^Y(Q,t,z) = z\mathbf{1} + t + O(z^{-1}). \tag{2}$$

The *extremal* specialization [13, (28) and (33)] sets $Q^d = 1$ for $d = k[L]$ with $[L]$ the class of a line in a fibre of $\mathbf{P}(V) \subset T \to Z$, sets $Q^d = 0$ for all other d, and replaces the parameter by

$$\bar{\mathbf{t}} = t + \ln(q)c_1(T), \qquad t \in H^*(Z). \tag{3}$$

In the specialized variables (2) reads $J^T(q,t,z) = z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$, and the prefactor $z\, e^{t/z} q^{c_1(T)/z} = z\, e^{\bar{\mathbf{t}}/z}$ of (1) is exactly the $d = 0$ summand. Whether (1) is the J-function is therefore the question of whether the summands with $d \geq 1$ contribute in z-order ≥ 0.

Three statements of [13] carry a hypothesis on $\nu = r - s$, and the whole of Sections 5–9 depends on them through the chain recorded below. We list each with the inequality it states and the statements that use it.

Location in [13]	Hypothesis	Used by
Theorem 4.4	$r - s > 1$	(37); Theorem 4.6; (39)–(40)
Remark 4.5(3)	$r - s \leq 1$	nothing; a negative assertion
Section 7, before (64)	$r - s > 1$	(64); Proposition 7.5 and its corollaries

The first row is the load-bearing one. [13, Theorem 4.6] deduces the presentation

$$QH^*_{\mathrm{ext}}(T) = H^*(Z)[H][[q]] \Big/ \Big\langle \textstyle\prod_{i=1}^r (\rho_i + H) = q^{r-s} \prod_{j=1}^s (\sigma_j - H) \Big\rangle, \tag{4}$$

which is [13, (37)], from (1) by exhibiting a quantum differential operator annihilating J^T and quoting [6, Theorem 10.3.1] and [12, Lemma 2]. That argument uses $r - s > 1$ only through [13, Theorem 4.4]. The spectral computations [13, Propositions 5.1 and 5.2, Corollary 5.3] then use (4) and the ring homomorphism of [13, Theorem 3.3], and carry no hypothesis on ν of their own; the inequality in [13, Theorem 1.2] is inherited from [13, Theorem 4.4]. Likewise, the modified extremal J-function [13, (40)] used throughout Sections 7–9 is derived from (1), and [13, Proposition 9.1] uses (4) again. Establishing (1) at $\nu = 1$ therefore unblocks all of this at once.

The second row asserts the opposite of what we prove, and is used nowhere: [13, Remark 4.5(3)] is explicitly marked as not needed, and the same is true of its counterpart [13, Remark 4.7] for T'. What we do use from it is its positive half, the statement that (1) lies on Givental's overruled Lagrangian cone of T for $r - s \leq 1$, which the authors obtain from [4], [5] and [13, Lemma 9.6].

The third row governs the aperture of the Barnes expansion only. Nothing in Sections 8 and 9 of [13] constrains ν: the sector [13, (78)] of [13, Proposition 8.2] comes from [13, Theorem A.2], whose hypothesis $|\arg t| < (\frac{\nu}{2} + 1)\pi$ carries no ϵ and is valid at $\nu = 1$; and the reductions of [13, Sections 9.1–9.4] — weak-to-strong in the local model, the reduction to the split case, the cohomological reduction to the local model, and the comparison of the Fourier–Mukai transform with the local model — are stated and proved with $r > s$ as the only inequality.

3 J-normalization at discrepancy one

By the splitting principle of [13, Corollary 4.3], we may pull back injectively to a common flag bundle over Z on which both V and V' have filtrations by line bundles. The classes ρ_i and σ_j then exist individually; we retain the same notation after pullback. Write

$$A = H^*(\mathbf{P}(V)) \cong H^*(Z)[H]/\left\langle \prod_{i=1}^{r}(\rho_i + H)\right\rangle$$

for the coefficient ring of (1). It is a finite-dimensional graded $\mathbf{C}$-algebra whose elements of positive degree are nilpotent. Each factor of the denominator of (1) has the form $mz + a$ with $m \geq 1$ and $a = \rho_i + H \in A$, so

$$\frac{1}{mz+a} = \frac{1}{mz}\sum_{l\geq 0}\left(\frac{-a}{mz}\right)^l \in z^{-1}A[z^{-1}]$$

is a finite sum, and (1) is an element of $zA[z^{-1}][[q]]$. By the *z-order* of such an element we mean the largest n with a nonzero z^n coefficient. Since the pullback is injective coefficientwise, the resulting order bound descends to the original cohomology ring; its symmetric expression is independent of the splitting.

Lemma 3.1. *For every $d \geq 1$, the degree-d summand of* (1) *has z-order at most*

$$1 + s(d-1) - rd = 1 - s - \nu d.$$

Proof. The numerator $\prod_{j=1}^{s}\prod_{m=0}^{d-1}(\sigma_j - H - mz)$ has sd factors. The s factors with $m = 0$ are $\sigma_j - H$ and contain no z; each of the remaining $s(d-1)$ factors is linear in z. So the numerator is a polynomial in z over A of degree at most $s(d-1)$. The denominator $\prod_{i=1}^{r}\prod_{m=1}^{d}(\rho_i + H + mz)$ has rd factors, all with $m \geq 1$, so by the expansion above its inverse lies in $z^{-rd}A[z^{-1}]$. Multiplying by the leading factor z of (1) adds one. □

The two hypotheses of Theorem 1.1 enter here and nowhere else.

Corollary 3.2. *If $\nu = 1$ and $s \geq 1$, then every summand of* (1) *with $d \geq 1$ has z-order at most $1 - s - d \leq -1$, and hence*

$$z\, e^{t/z} q^{c_1(T)/z} \sum_{d\geq 0} q^{(r-s)d}\, \frac{\prod_{j=1}^{s}\prod_{m=0}^{d-1}(\sigma_j - H - mz)}{\prod_{i=1}^{r}\prod_{m=1}^{d}(\rho_i + H + mz)} = z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1}),$$

with $\bar{\mathbf{t}} = t + \ln(q)c_1(T)$ as in (3).

Proof. The bound of Lemma 3.1 is $1 - s - d$, which is at most -1 for $d \geq 1$ and $s \geq 1$. The $d = 0$ summand is $z\, e^{t/z} q^{c_1(T)/z} = z\, e^{\bar{\mathbf{t}}/z} = z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$. □

Remark 3.3. Both hypotheses are sharp. For $\nu = 1$ and $s = 0$, so $(r, s) = (1, 0)$, the bound of Lemma 3.1 is $1 - d$ and is attained by the formal $d = 1$ term of the hypergeometric expression:

$$z \cdot \frac{q}{\rho_1 + H + z} = q\mathbf{1} - \frac{q(\rho_1 + H)}{z} + \cdots,$$

Its z^0 coefficient is $q\mathbf{1}$, so this formal expression is not J-normalized. Geometrically $\mathbf{P}(V) = Z$ has point fibres and no line class $[L]$, so $(1, 0)$ is a degenerate endpoint of the extremal-line setup, not a further standard-flip case repaired here. For $\nu = 0$ the bound is the d-independent value $1 - s$ and the sum is not q-adically convergent, which is why $r > s$ is assumed throughout [13].

To convert the normalization into an identification we use the standard characterization of the J-function as a slice of Givental's cone. Let $\mathcal{L}_T$ denote the overruled Lagrangian cone of the genus-zero Gromov–Witten theory of T, in the extremal specialization (3). The specialization is legitimate for the cone formalism because q carries positive weight: by [13, (33)] the curve class $k[L]$ contributes $q^{(r-s)k}$, and $r > s$, so $\mathcal{L}_T$ is defined over $\mathbf{C}[[q]]$ and (1) is a q-adically convergent point of the ambient space.

Lemma 3.4 (J-slice uniqueness; [8], [5])**.** *Let $f \in \mathcal{L}_T$ be of the form $f = z\mathbf{1} + \tau + O(z^{-1})$ for some τ in the parameter space. Then $f = J^T(\tau, z)$.*

This is the mechanism behind every mirror theorem of Givental type: a point of the cone with a prescribed z^0 coefficient is the J-function at that coefficient, and the mirror map is precisely the discrepancy between the prescribed z^0 coefficient and the parameter one started with.

Proposition 3.5. *Let $r = s + 1$ with $s \geq 1$. Then the right-hand side of* (1) *equals the extremal J-function $J^T(q, t, z)$ of the local model, in the variables and completion of [13, Theorem 4.4].*

Proof. By [13, Remark 4.5(3)] the right-hand side of (1) lies on $\mathcal{L}_T$ whenever $r - s \leq 1$; the authors deduce this from [4], [5] and [13, Lemma 9.6]. By Corollary 3.2 it has the form $z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$. Lemma 3.4 with $\tau = \bar{\mathbf{t}}$ gives the claim. ☐

Corollary 3.6. *Let $r = s + 1$ with $s \geq 1$. Then the presentation* (4) *of [13, Theorem 4.6] holds, the modified extremal J-function [13, (39)–(40)] of T is as printed, and [13, Theorem 1.2] holds: after the extremal specialization, $c_1(X) \star_q -$ on $QH^*(X)$ has eigenvalue* 0 *with multiplicity* $\operatorname{rank} H^*(X')$ *and the single nonzero eigenvalue*

$$\lambda_0(q) = e^{-\pi\sqrt{-1}\,s} q = (-1)^s q$$

with multiplicity $\operatorname{rank} H^*(Z)$*.*

Proof. The proof of [13, Theorem 4.6] specializes the parameter t of [13, Theorem 4.4], exhibits a differential operator annihilating J^T, and concludes with [6, Theorem 10.3.1] and [12, Lemma 2]; it uses $r - s > 1$ only to know that (1) is J^T. Proposition 3.5 supplies that, so the proof applies verbatim, and [13, (39)–(40)] follow by the same substitution as in [13, Section 4.3].

For the spectrum, [13, Propositions 5.1 and 5.2] concern the abstract ring (4) and its subspaces and assume only $r > s$, and [13, Corollary 5.3] combines them with the ring homomorphism $i^*\colon QH^*_{\text{ext}}(X) \to QH^*_{\text{ext}}(T)$ of [13, Theorem 3.3]. Once (4) is available at $\nu = 1$, that argument is unchanged. Its output is $r - s$ nonzero eigenvalues $\lambda_k(q) = (r-s)e^{-\pi\sqrt{-1}(2k+s)/(r-s)}q$ for $0 \leq k < r - s$; at $r - s = 1$ only $k = 0$ occurs, and $\lambda_0(q) = e^{-\pi\sqrt{-1}s}q$. ☐

Remark 3.7. The companion statement [13, (34)] for $\mathbf{P}(V)$ is the $s = 0$ case of the same expression, and Lemma 3.1 applied to it gives maximal z-order $1 - rd$. This is at most -1 for every $d \geq 1$ as soon as $r \geq 2$, which holds whenever $r = s + 1$ with $s \geq 1$. So [13, (34), (36) and (38)] are covered in the same range, by the same count.

Proposition 3.5 and Corollary 3.6 prove Theorem 1.1. They also settle the status of [13, Remark 4.5(3)]: under the standing assumption $r > s$ its range $r - s \leq 1$ is exactly $\nu = 1$, and by Corollary 3.2 and Remark 3.3 its negative assertion holds there only for $s = 0$.

4 The Meijer aperture at $\nu = 1$

The asymptotic analysis of [13, Section 7] rests on [13, Lemma 7.4], which writes the equivariant central-charge kernel κ_m as a cohomology-valued Meijer G-function,

$$\kappa_m = e^{-\pi\sqrt{-1}c_1(V')}\, G^{r,0}_{s,r}\left(\begin{matrix} 1-\sigma_1,\dots,1-\sigma_s \\ \rho_1,\dots,\rho_r \end{matrix}\,\middle|\, t_m\right), \qquad t_m := e^{-\pi\sqrt{-1}(2m+s)}\Big(\frac{q}{z}\Big)^{r-s}. \tag{5}$$

In the notation of [13, Appendix A] the lower and upper index counts of this G-function are s and r, so the parameter governing the asymptotic expansion is

$$\nu = r - s, \tag{6}$$

which is the discrepancy of the flip. (The appendix of [13] names the index counts p and q; we write s and r to keep q for the extremal variable.)

[13, Theorem A.1] is Barnes' asymptotic expansion [2] in the form given by Meijer [10] and Luke [9]. Its hypothesis is stated there as

$$|\arg t| < (\nu+\epsilon)\pi, \qquad \epsilon = 1 \text{ if } \nu > 1, \qquad \epsilon = \tfrac{1}{2} \text{ if } \nu = 1. \tag{7}$$

The ϵ of (7) is the whole issue. The expansion printed in [13, Section 7] after [13, Lemma 7.4] is introduced by the sentence "For $r - s > 1$", and the sector recorded with it,

$$\left|\arg\Big(\frac{z}{q}\Big) + \frac{(2m+s)\pi}{r-s}\right| < \Big(1 + \frac{1}{r-s}\Big)\pi, \tag{8}$$

which is [13, (64)], is exactly (7) with $\epsilon = 1$ transported through (5).

Lemma 4.1. *Let $\nu = 1$. Applied to (5), [13, Theorem A.1] gives the asymptotic expansion of [13, Proposition 7.5] on*

$$\left|\arg\Big(\frac{z}{q}\Big) + (2m+s)\pi\right| < \frac{3\pi}{2}, \tag{9}$$

. The sector obtained by substituting $r - s = 1$ into (8) is $\big|\arg(z/q) + (2m+s)\pi\big| < 2\pi$, wider by $\pi/2$ on each side.

Proof. From (5), $\arg t_m = -(2m+s)\pi - \nu\arg(z/q)$. At $\nu = 1$ the hypothesis (7) reads $|\arg t_m| < (1+\frac{1}{2})\pi$, which is (9). Substituting $r - s = 1$ into (8) gives $(1+1)\pi = 2\pi$, the value of $(\nu+\epsilon)\pi$ at $\epsilon = 1$. $\square$

The correction is one-directional: (9) is smaller than the naive specialization of (8), so a repair must show that what remains is still enough. The other sector in play is unaffected.

Lemma 4.2. *[13, Proposition 8.2] and its sector*

$$\left|\arg\Big(\frac{z}{q}\Big) - \frac{(1-s)\pi}{r-s}\right| < \frac{\pi}{2} + \frac{\pi}{r-s}, \tag{10}$$

which is [13, (78)], are valid at $\nu = 1$, where (10) reads $\big|\arg(z/q) - (1-s)\pi\big| < \frac{3\pi}{2}$.

Proof. [13, Proposition 8.2] is deduced from [13, (76)], which applies [13, Theorem A.2] to a $G^{r,1}_{s,r}$ function of the same argument. The hypothesis of [13, Theorem A.2] is $|\arg t| < (\frac{\nu}{2}+1)\pi$; it involves no ϵ and no restriction on ν. Transporting it through the same substitution as above gives (10) for every $\nu \geq 1$, and the displayed form at $\nu = 1$. $\square$

Proposition 4.3. *Let $r = s + 1$ with $s \geq 1$ and let $0 \leq k \leq 1$, so that the semiorthogonal decomposition [13, Theorem 1.1] has the single Z-component of index $m = -k$. The sectors* (9) *and* (10) *meet in the open sector*

$$\max\big((2k-s)\pi,\,(1-s)\pi\big) - \frac{3\pi}{2} < \arg\Big(\frac{z}{q}\Big) < \min\big((2k-s)\pi,\,(1-s)\pi\big) + \frac{3\pi}{2}, \qquad (11)$$

of opening 2π. It contains the eigenvalue ray $\arg(z/q) = (2k - s)\pi$ of [13, Theorem 1.4(1)] and the tame ray $\arg(z/q) = (1 - s)\pi$ of [13, Theorem 1.4(2)].

Proof. With $m = -k$, sector (9) is the open interval of half-width $3\pi/2$ about $(2k - s)\pi$, and sector (10) is the open interval of half-width $3\pi/2$ about $(1 - s)\pi$. For $k \in \{0, 1\}$ the two centres are at distance $|2k - 1|\pi = \pi$, so the intervals meet in an interval of length $3\pi - \pi = 2\pi$, namely (11). Since $\pi < 3\pi/2$, each centre lies in the other interval, so both centres lie in the intersection; they are the two rays named in the statement, because at $r - s = 1$ the rays $-(2m + s)\pi/(r - s)$ and $(1 - s)\pi/(r - s)$ of [13, Theorem 1.4] are $(2k - s)\pi$ and $(1 - s)\pi$. □

Lemmas 4.1 and 4.2 and Proposition 4.3 prove Theorem 1.2.

Remark 4.4. For a blow-up, $s = 1$, the numbers are: (9) with $k = m = 0$ is $-\frac{5\pi}{2} < \arg(z/q) < \frac{\pi}{2}$; (10) is $|\arg(z/q)| < \frac{3\pi}{2}$; and (11) is $-\frac{3\pi}{2} < \arg(z/q) < \frac{\pi}{2}$, containing the tame ray 0 and the eigenvalue ray $-\pi$. The codimension-two blow-up is $(r, s) = (2, 1)$, where $\lambda_0(q) = -q$ and the exponential factor of the expansion of [13, Proposition 7.5] is $\exp(-\lambda_0(q)/z) = \exp(q/z)$.

Remark 4.5. The opening 2π of (11) exceeds π, the width beyond which an asymptotic expansion of level one determines its flat section uniquely; see [1]. The oriented comparison of the nonzero-eigenvalue and tame expansions carried out in [13, Section 9], which needs both expansions to be valid simultaneously on a sector containing both rays, is therefore available at $\nu = 1$. In particular the common-sector condition of [13, Remark 1.6], namely $\frac{1}{4}(r - s - 6) < k < \frac{3}{4}(r - s + 2)$, which at $r - s = 1$ reads $-\frac{5}{4} < k < \frac{9}{4}$ and admits both $k = 0$ and $k = 1$, holds at $\nu = 1$ for the corrected sector (9) and not only for (8).

5 What is repaired, and what is not

Theorem 5.1. *Let $X \dashrightarrow X'$ be a standard flip with $r = s + 1$ and $s \geq 1$. Then [13, Theorems 1.2, 1.4, 9.9 and 9.14] and [13, Corollary 1.5] hold for X. In particular they hold for the blow-up of a smooth projective variety along a smooth centre of codimension two, the case $(r, s) = (2, 1)$.*

Proof. Corollary 3.6 gives (4) and [13, Theorem 1.2], and makes [13, (39)–(40)] available, so the modified extremal J-function used from [13, Section 7] onward is the one the argument assumes. [13, Proposition 7.5], [13, Corollary 7.6] and [13, Proposition 7.8] then hold with (8) replaced by (9), by Lemma 4.1; [13, Propositions 8.2 and 8.3] and [13, Corollary 8.4] hold as printed, by Lemma 4.2.

The reductions of [13, Sections 9.1–9.4] are then unchanged. [13, Proposition 9.1] uses the presentation (4) and the extremal Dubrovin connection; [13, Proposition 9.2] combines it with [13, Propositions 7.8 and 8.4]; [13, Corollary 9.3] takes a partial nonequivariant limit; [13, Section 9.2] removes the splitting assumption on V, V', giving [13, Corollary 9.7]; [13, Section 9.3] passes from the local model to X, giving [13, Theorem 9.9]; and [13, Section 9.4] compares the Fourier–Mukai transform with the local model, giving [13, Theorem 9.14]. None of these steps constrains ν beyond $r > s$. The two halves of [13, Theorem 1.4] are

[13, Theorems 9.9 and 9.14], and [13, Corollary 1.5] is the translation of [13, Theorem 1.4] through the functors [13, (3)]. Proposition 4.3 supplies the sector on which both expansions used in that translation are valid. □

Three boundaries should be stated explicitly.

First, $(r, s) = (1, 0)$ is not covered. By Remark 3.3 the formal expression (1) has a z-order-zero term there and is not J-normalized. This is the rank-one projective-bundle endpoint, where $\mathbf{P}(V) = Z$, $X' = \varnothing$, and the point fibres contain no line class $[L]$. It therefore lies outside both the normalization argument and the geometric extremal-line setup. All genuine discrepancy-one standard flips have $s \geq 1$ and are covered by Theorem 5.1.

Second, everything above lives on the extremal slice. The Novikov parameter is restricted to multiples of the class $[L]$ of a line in a fibre of $\mathbf{P}(V)$, all other Novikov variables are zero, and the parameter is $\bar{\mathbf{t}} = t + \ln(q)c_1(T)$ with t pulled back from $H^*(Z)$; this is the specialization in which [13, Theorems 1.2 and 1.4] are stated, and no statement here concerns the unrestricted quantum product or a comparison across the full Novikov ring.

Third, the two corrections are local to the two hypotheses named in Section 2. For $\nu > 1$ nothing changes: the argument of [13] applies as printed, with (8) supplied by the $\epsilon = 1$ branch of [13, Theorem A.1]. Every other ingredient of [13] — the ring homomorphism to the local model, the spectral computation in the abstract presentation, the deformation to the normal cone, the Fourier–Mukai comparison — is used exactly as it stands.

The count in Lemma 3.1 is what makes the discrepancy-one range behave like the higher-discrepancy one. The s numerator factors $\sigma_j - H$ at $m = 0$, free of z, are worth exactly s powers, and at $\nu = 1$ they are the entire margin: the maximal order $1 - s - \nu d$ sits at -1 precisely when $s = 1$ and $d = 1$, and rises to 0 the moment s drops to zero. So the boundary between the identification $I = J$ and a nontrivial mirror map at $\nu = 1$ is not the codimension of the centre but the presence of at least one factor in V'.

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